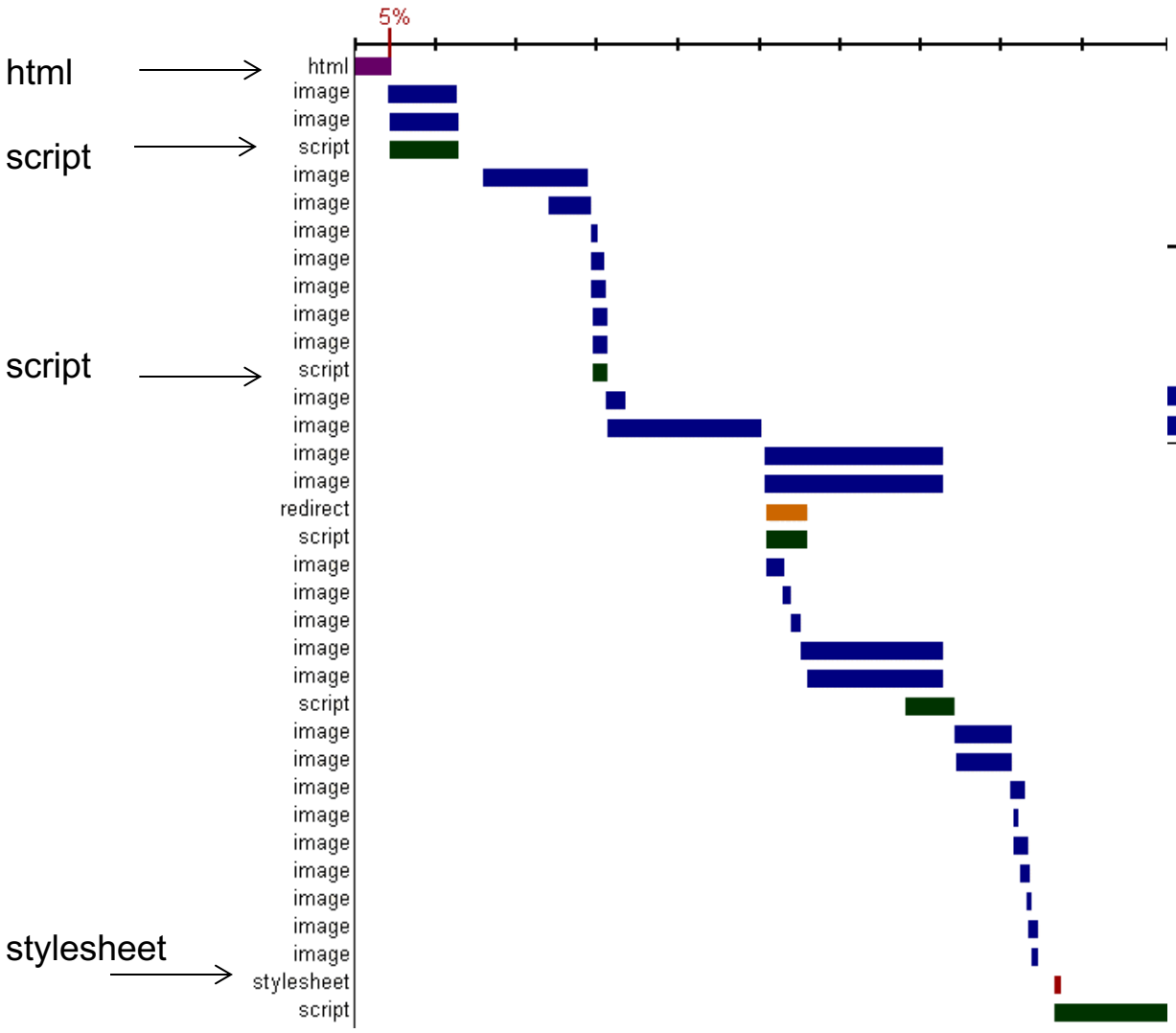
High Performance Web Sites

Much of the material derives from Steve Souders (SpeedCurve) and Tenni Theurer (Visa), 2006 research at Yahoo!

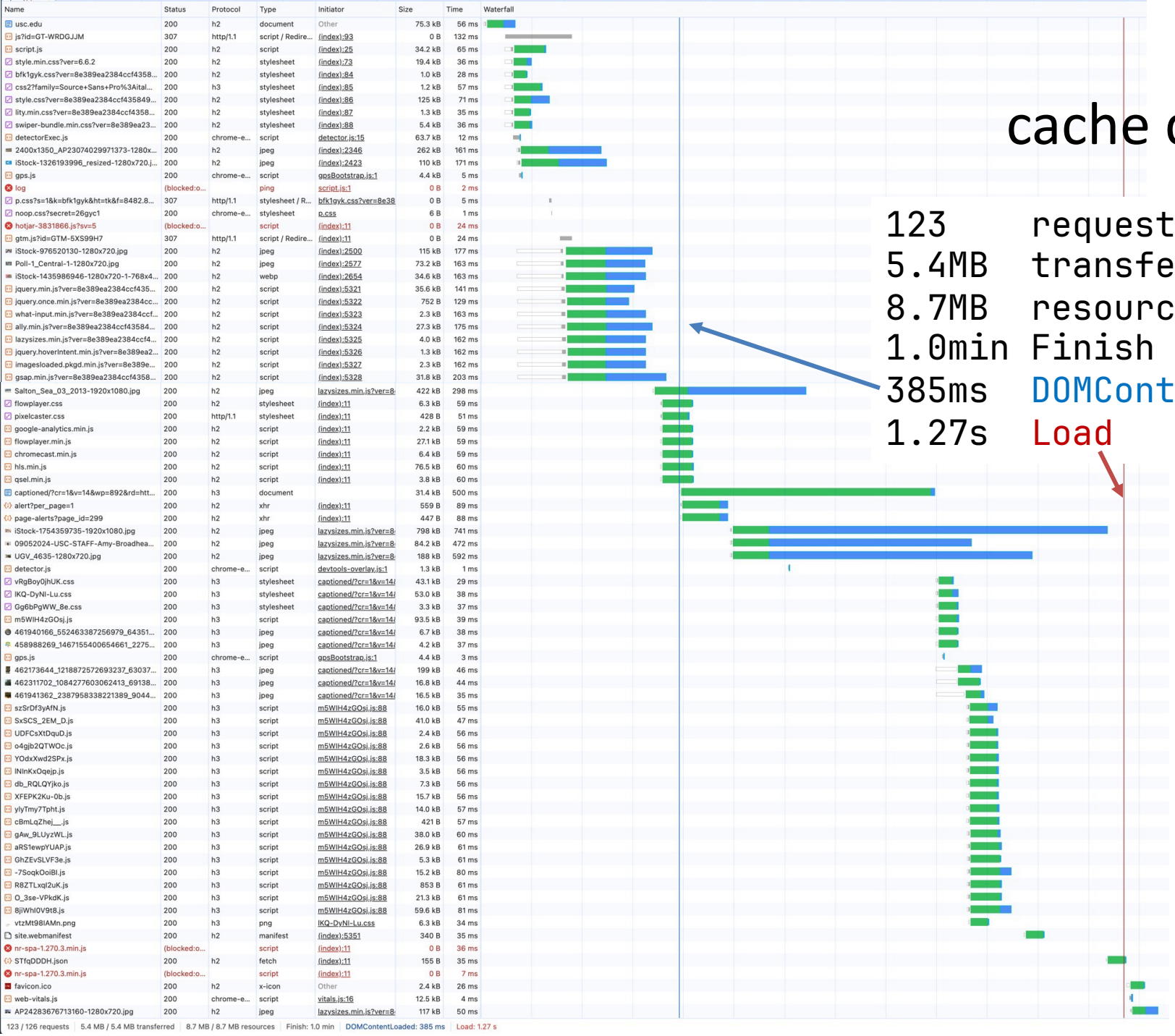
The Importance of Front-End Performance



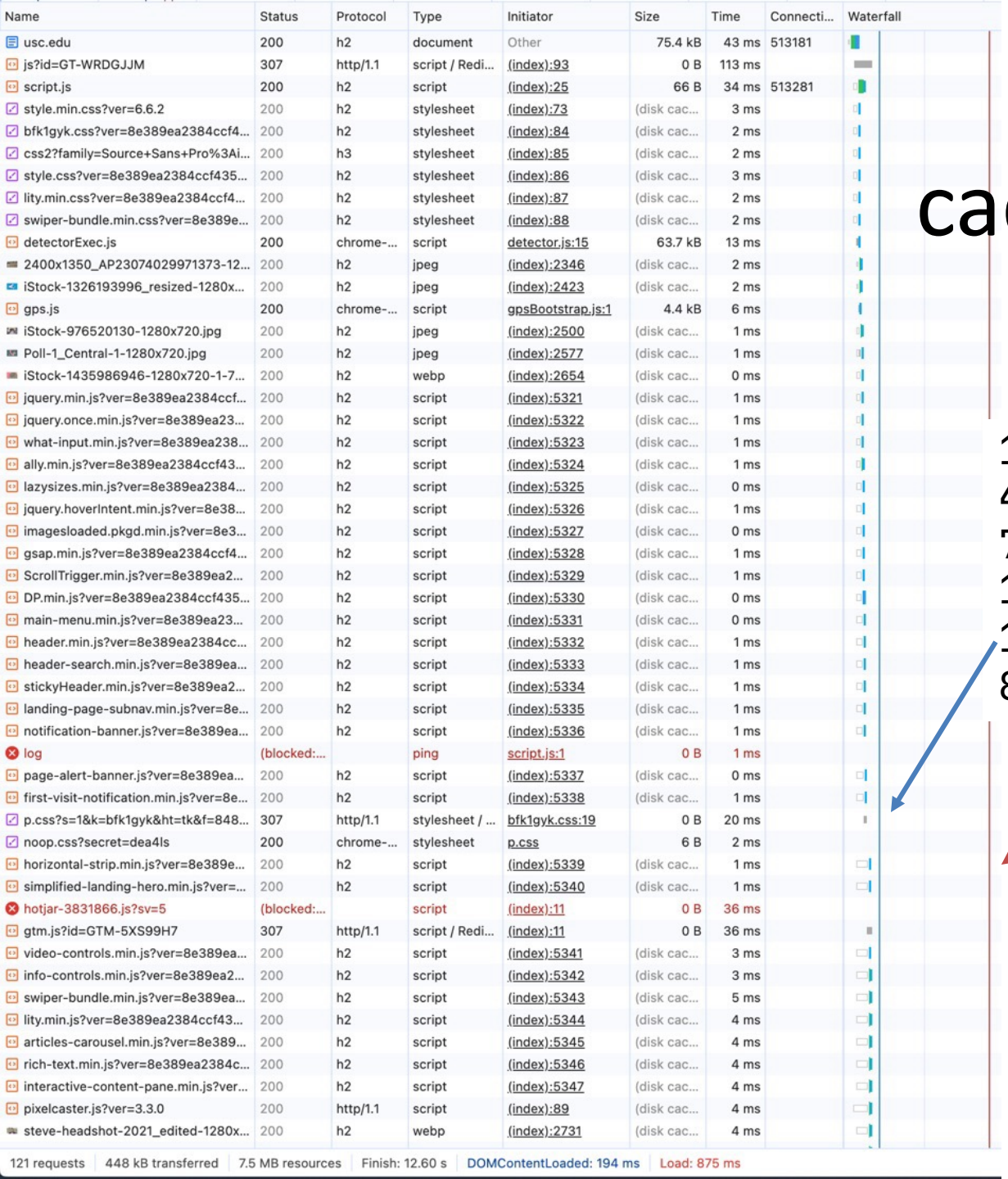
The graph shows the time taken to download an html file and various associated image files, JavaScript files, and CSS files. Note the time to download the html is a very small percentage of the total time.

(See browser tools Network tabs showing “timeline”)

usc.edu
cache disabled



usc.edu cache enabled



Name	Status	Protocol	Type	Initiator	Size	Time	Connecti...	Waterfall
usc.edu	200	h2	document	Other	75.4 kB	43 ms	513181	
js?id=GT-WRDGJJM	307	http/1.1	script / Redi...	(index):93	0 B	113 ms		
script.js	200	h2	script	(index):25	66 B	34 ms	513281	
style.min.css?ver=6.6.2	200	h2	stylesheet	(index):73	(disk cac...	3 ms		
bfk1gyk.css?ver=8e389ea2384ccf4...	200	h2	stylesheet	(index):84	(disk cac...	2 ms		
css2?family=Source+Sans+Pro%3Ai...	200	h3	stylesheet	(index):85	(disk cac...	2 ms		
style.css?ver=8e389ea2384ccf435...	200	h2	stylesheet	(index):86	(disk cac...	3 ms		
lity.min.css?ver=8e389ea2384ccf4...	200	h2	stylesheet	(index):87	(disk cac...	2 ms		
swiper-bundle.min.css?ver=8e389e...	200	h2	stylesheet	(index):88	(disk cac...	2 ms		
detectorExec.js	200	chrome-...	script	detector.js:15	63.7 kB	13 ms		
2400x1350_AP23074029971373-12...	200	h2	jpeg	(index):2346	(disk cac...	2 ms		
iStock-1326193996_resized-1280x...	200	h2	jpeg	(index):2423	(disk cac...	2 ms		
gps.js	200	chrome-...	script	gpsBootstrap.js:1	4.4 kB	6 ms		
iStock-976520130-1280x720.jpg	200	h2	jpeg	(index):2500	(disk cac...	1 ms		
Poll-1_Central-1-1280x720.jpg	200	h2	jpeg	(index):2577	(disk cac...	1 ms		
iStock-1435986946-1280x720-1-7...	200	h2	webp	(index):2654	(disk cac...	0 ms		
jquery.min.js?ver=8e389ea2384ccf...	200	h2	script	(index):5321	(disk cac...	1 ms		
jquery.once.min.js?ver=8e389ea23...	200	h2	script	(index):5322	(disk cac...	1 ms		
what-input.min.js?ver=8e389ea238...	200	h2	script	(index):5323	(disk cac...	1 ms		
ally.min.js?ver=8e389ea2384ccf43...	200	h2	script	(index):5324	(disk cac...	1 ms		
lazysizes.min.js?ver=8e389ea2384...	200	h2	script	(index):5325	(disk cac...	0 ms		
jquery.hoverIntent.min.js?ver=8e38...	200	h2	script	(index):5326	(disk cac...	1 ms		
imagesloaded.pkgd.min.js?ver=8e3...	200	h2	script	(index):5327	(disk cac...	0 ms		
gsap.min.js?ver=8e389ea2384ccf4...	200	h2	script	(index):5328	(disk cac...	1 ms		
ScrollTrigger.min.js?ver=8e389ea2...	200	h2	script	(index):5329	(disk cac...	1 ms		
DP.min.js?ver=8e389ea2384ccf435...	200	h2	script	(index):5330	(disk cac...	0 ms		
main-menu.min.js?ver=8e389ea23...	200	h2	script	(index):5331	(disk cac...	0 ms		
header.min.js?ver=8e389ea2384cc...	200	h2	script	(index):5332	(disk cac...	1 ms		
header-search.min.js?ver=8e389ea...	200	h2	script	(index):5333	(disk cac...	1 ms		
stickyHeader.min.js?ver=8e389ea2...	200	h2	script	(index):5334	(disk cac...	1 ms		
landing-page-subnav.min.js?ver=8e...	200	h2	script	(index):5335	(disk cac...	1 ms		
notification-banner.js?ver=8e389ea...	200	h2	script	(index):5336	(disk cac...	1 ms		
log	(blocked:...		ping	script.js:1	0 B	1 ms		
page-alert-banner.js?ver=8e389ea...	200	h2	script	(index):5337	(disk cac...	0 ms		
first-visit-notification.min.js?ver=8e...	200	h2	script	(index):5338	(disk cac...	1 ms		
p.css?s=1&k=bfk1gyk&ht=tk&f=848...	307	http/1.1	stylesheet / ...	bfk1gyk.css:19	0 B	20 ms		
noop.css?secret=dea4ls	200	chrome-...	stylesheet	p.css	6 B	2 ms		
horizontal-strip.min.js?ver=8e389e...	200	h2	script	(index):5339	(disk cac...	1 ms		
simplified-landing-hero.min.js?ver=...	200	h2	script	(index):5340	(disk cac...	1 ms		
hotjar-3831866.js?sv=5	(blocked:...		script	(index):11	0 B	36 ms		
gtm.js?id=GTM-5XS99H7	307	http/1.1	script / Redi...	(index):11	0 B	36 ms		
video-controls.min.js?ver=8e389ea...	200	h2	script	(index):5341	(disk cac...	3 ms		
info-controls.min.js?ver=8e389ea2...	200	h2	script	(index):5342	(disk cac...	3 ms		
swiper-bundle.min.js?ver=8e389ea...	200	h2	script	(index):5343	(disk cac...	5 ms		
lity.min.js?ver=8e389ea2384ccf43...	200	h2	script	(index):5344	(disk cac...	4 ms		
articles-carousel.min.js?ver=8e389...	200	h2	script	(index):5345	(disk cac...	4 ms		
rich-text.min.js?ver=8e389ea2384c...	200	h2	script	(index):5346	(disk cac...	4 ms		
interactive-content-pane.min.js?ver...	200	h2	script	(index):5347	(disk cac...	4 ms		
pixelcaster.js?ver=3.3.0	200	http/1.1	script	(index):89	(disk cac...	4 ms		
steve-headshot-2021_edited-1280x...	200	h2	webp	(index):2731	(disk cac...	4 ms		

121 requests | 448 kB transferred | 7.5 MB resources | Finish: 12.60 s | DOMContentLoaded: 194 ms | Load: 875 ms

```
121     requests
448kB   transferred
7.5MB   resources
12.60s  Finish
194ms   DOMContentLoaded
875ms   Load
```

Where Is The Most Of The Time Spent

A study of popular web pages and the time to download them showed that the vast majority of time is spent on the client side;

This is true even if the page has been cached

	Empty Cache	Full Cache
amazon.com	82%	86%
aol.com	94%	86%
cnn.com	81%	92%
ebay.com	98%	92%
google.com	86%	64%
msn.com	97%	95%
myspace.com	96%	86%
wikipedia.org	80%	88%
yahoo.com	95%	88%
youtube.com	97%	95%

percentage of time spent on the front-end

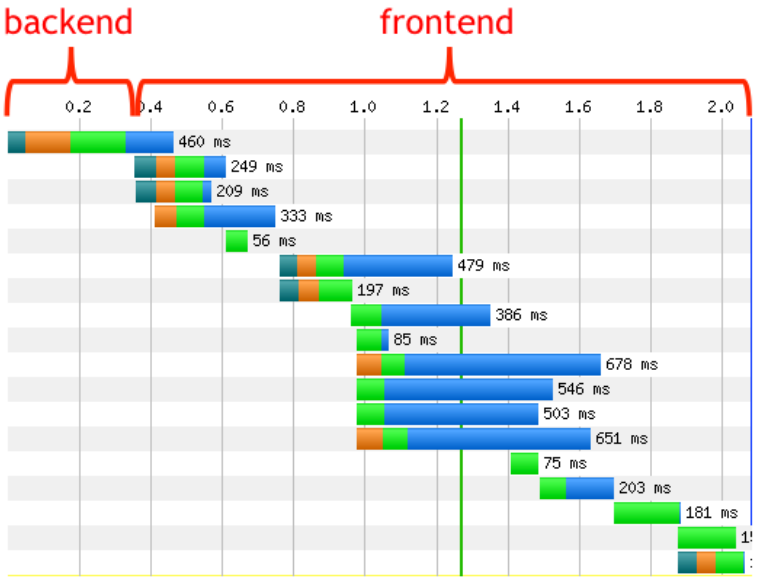
80/20 Performance Rule

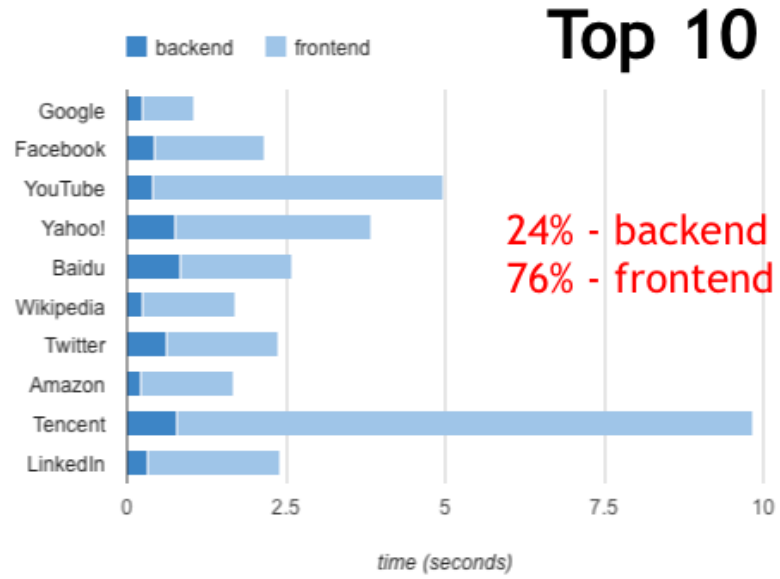
- Vilfredo Pareto, Italian economist, “Pareto Principle”:
  - “80% of the consequences come from 20% of the causes”

  https://www.investopedia.com/terms/p/paretoprinciple.asp
- Software Engineering: “80% of time spent in 20% of the code”
- Focus on the 20% that affects 80% of the end-user response time
- Web pages: 10% spent fetching HTML, 90% spent on fetching Images, scripts and stylesheets
- I.e. , start at the front-end

The Performance Golden Rule

- 80-90% of the end-user response time is spent on the front-end. So, start there.
  - Greater potential for improvement
  - Simpler to optimize
  - Proven to work





Browser networking

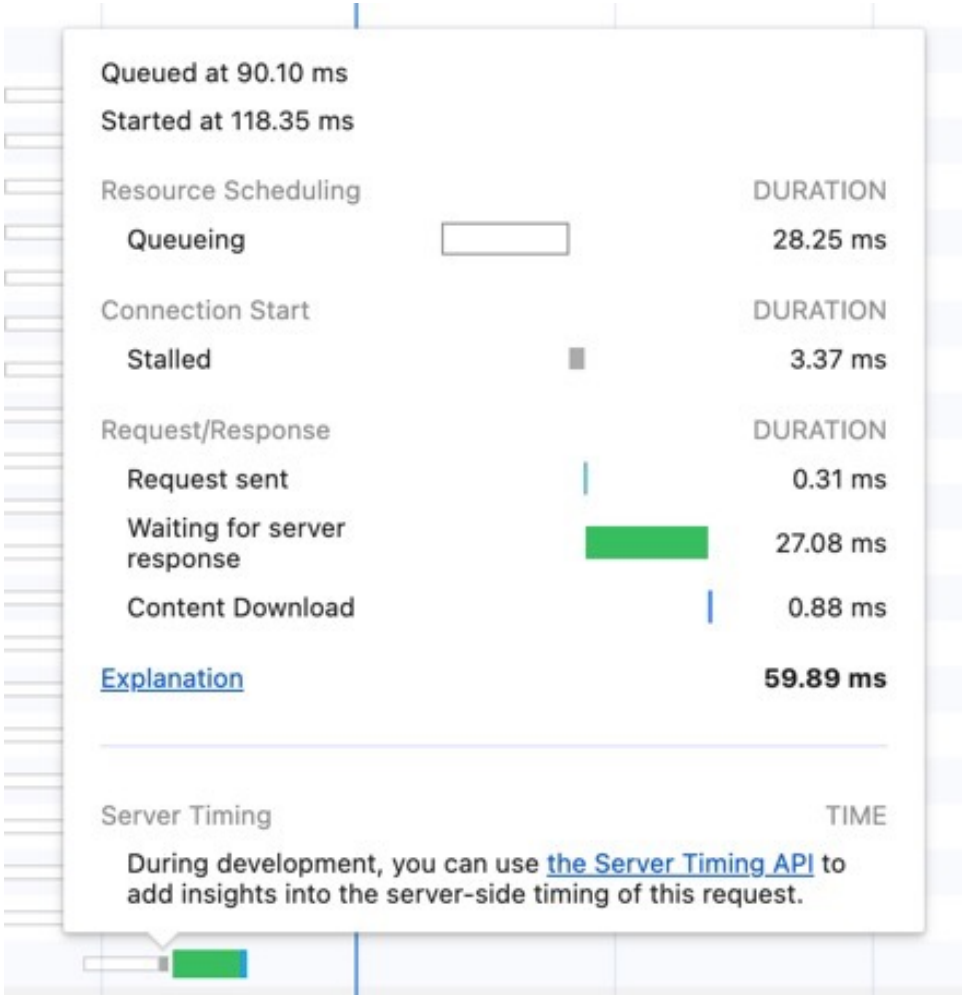
- Browsers typically allow up to 6-8 simultaneous TCP connections per domain.

- For each connection there are protocol limitations:
  - HTTP 1.0/1.1: no parallel requests over single connection.
    Browser will wait until current request finishes before sending next request
  - HTTP 2 (h2): can issue multiple (in theory unlimited, 100 in practice) requests
    over same connection without need to wait for previous requests to finish

- The HTTP 1.1 protocol states that single-user clients should not maintain more than two connections with any server or proxy.

Browser	Max # connections (2020)	
	per domain	total
Chrome 81	6	256
Edge 18	12	>1000
Firefox 68	6+3	>1000
Safari 13	6	>1000

Request timings



Queueing. The browser queues requests before connection start and when:

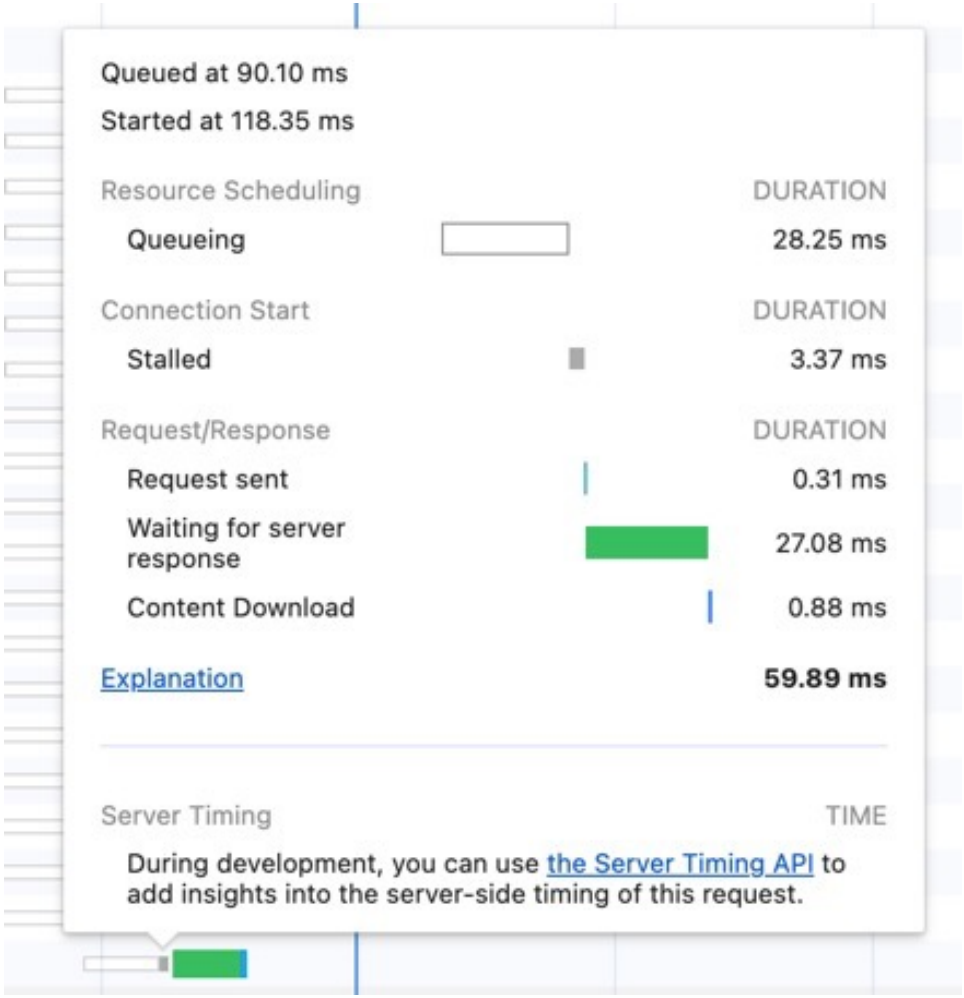
- There are higher priority requests.
- There are already six TCP connections open for this origin, which is the limit. Applies to HTTP/1.0 and HTTP/1.1 only.
- The browser is briefly allocating space in the disk cache.

Stalled. The request could be stalled after connection start for any of the reasons described in **Queueing**.

DNS Lookup. The browser is resolving the request's IP address.

Initial connection. TCP handshakes or retries and negotiating an SSL/TLS.

Request timings (cont'd)



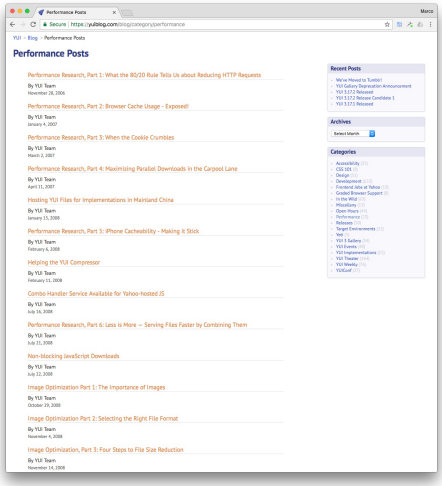
Proxy negotiation. The browser is negotiating the request with a proxy server.

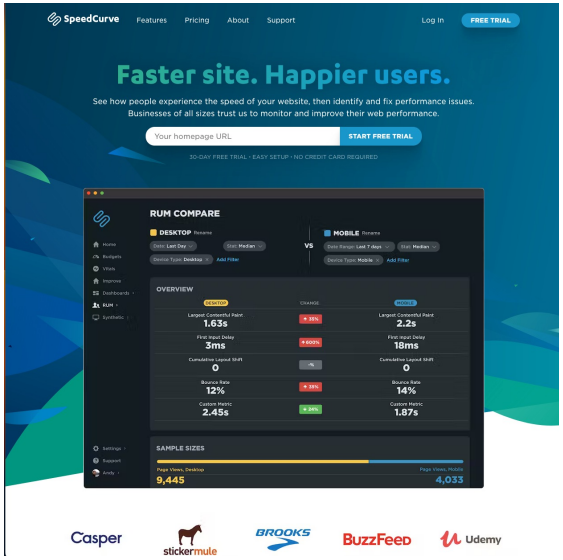
Request sent. The request is being sent.

Waiting (TTFB). The browser is waiting for the first byte of a response.
TTFB stands for Time To First Byte. This timing includes 1 round trip of latency and the time the server took to prepare the response.

Content Download. Downloading whole response (waiting for the last byte) + processing/parsing the data.

Yahoo Interface / Engineering Blogs





Most of the work on how to improve the performance of Web sites was done by **Steve Souders** and **Tenni Theurer** at Yahoo.com. Original blog "Performance " presentation (2006-2011):
https://stevesouders.com/docs/rich-web-experience-souders-theurer.ppt
https://slideplayer.com/slide/678200/

Steve Souders was Head Performance Engineer at Google, Chief Performance at Yahoo and Fastly, a CDN, and now is at **SpeedCurve**.

See his work at: http://stevesouders.com

"SpeedCurve provides insight into the interaction between performance and design to help companies deliver fast and engaging user experiences."
https://www.speedcurve.com

See Blog at:
https://www.stevesouders.com/blog/

Tenni Theurer moved to VISA.

The importance of cache - Experiment

- An "empty cache" means the browser bypasses the disk cache and has to request all the components to load the page.
- A "full cache" means all (or at least most) of the components are found in the disk cache and the corresponding HTTP requests are avoided
- Experiment: Try to determine what the percentage of people is who load a home page when there are no elements of the page in the user's cache?
- Solution: add a new image (a pixel) to your page, e.g.

```
<img src="image/blank.gif" height="1" width="1"/>
```

- With the following response headers:

```
Expires: Tue, 1 Mar 2022 20:00:00 GMT       (an earlier date than today)
Last-Modified: Tue, 22 Mar 2022 12:30:00 GMT  (today's date)
```

- The Expires makes sure the page is not cached; the Last-Modified makes sure the server will have to check if blank.gif has changed
- Requests from a browser will produce one of these response status codes:
  - `200` – the server is sending back the image implying the browser does not have the image in its cache
  - `304` – the browser has the image in its cache, and the server responds saying it has not been modified
- Compute the following numbers:
  - Percentage of users who view with an empty cache ::=
    - (# unique users with at least one 200 response)/(total # unique users)
  - Percentage of page views that are done with an empty cache ::=
    - (total # of 200 responses) / (# of 200 + # of 304 responses)

The importance of cache – Experiment (2)

- See: https://www.stevesouders.com/blog/2008/04/30/high-performance-web-sites-part-2/

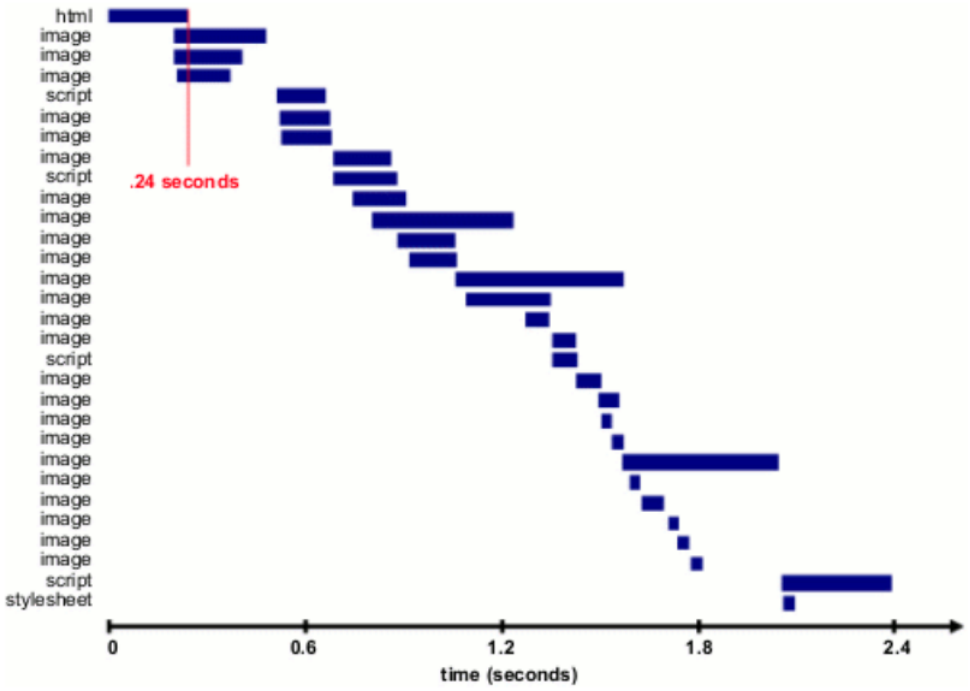


Figure 1. Loading http://www.yahoo.com with an empty cache

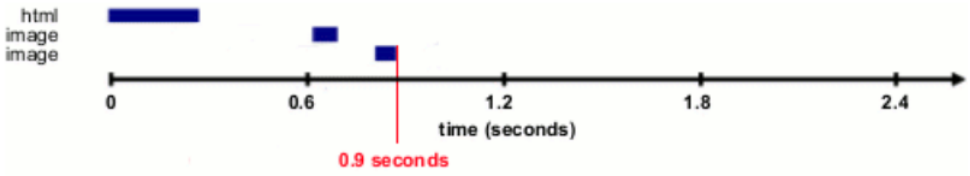


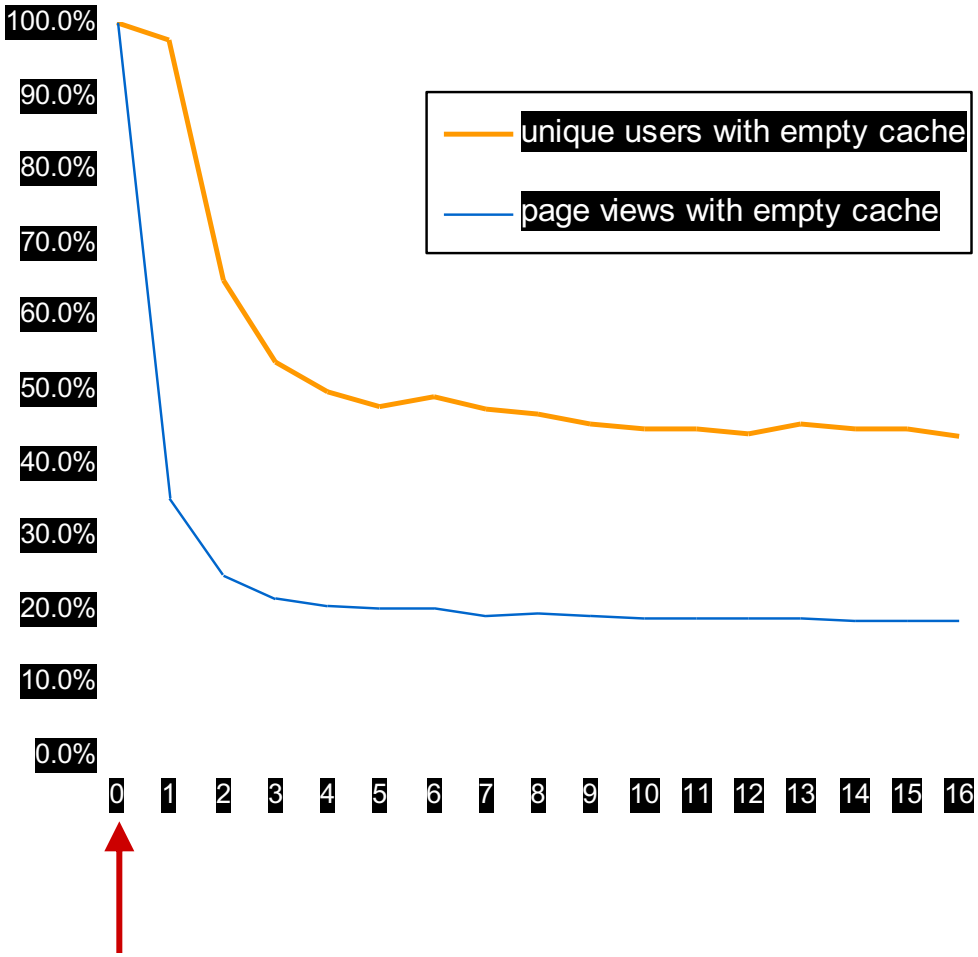
Figure 2. Loading http://www.yahoo.com with a full cache

Surprising Results Lessons:

The empty cache user is more prevalent than one might think

users with empty cache

40-60%



page views with empty cache

~20%

On the first day no one has the images cached, but over time more people have the image until a steady state is reached; Result: 40-60% of yahoo users have an empty cache experience and 20% of page views are done with an empty cache. Even if your assets are optimized for maximum caching, **there are a significant number of users that will always have an empty cache.**

Impact of Cookies on Response Time

Cookie Size	**time**	**Delta**
0 bytes	78ms	0ms
500 bytes	79ms	+1 ms
1000 bytes	94ms	+16ms
1500 bytes	109ms	+31ms
2000 bytes	125ms	+47ms
2500 bytes	141ms	+63ms
3000 bytes	156ms	+78ms

ms = millisecond (at ~800kbps, DSL speed)

(Note: today's cookies are much bigger in size, as big as megabytes)

Analysis of Cookie Sizes Across the Web

Website total Cookie Size (2007)

Amazon	60 bytes
Google	72 bytes
Yahoo	122 bytes
Cnn	184 bytes
Youtube	218 bytes
msn	268 bytes
eBay	331 bytes
MySpace	500 bytes

Lessons

- eliminate unnecessary cookies
- keep cookie sizes low
- set cookies at appropriate domain level
- set Expires date appropriately

an earlier date or none removes the cookie sooner

- Unfortunately, today's cookie sizes are **much, much bigger**

The “initial” 14 Rules

outdated

1. Make fewer HTTP requests
2. Use a CDN (content distribution network)
3. Add an Expires header
4. Gzip components
5. Put stylesheets at the top
6. Move scripts to the bottom
7. Avoid CSS expressions
8. Make JS and CSS external
9. Reduce DNS lookups
10. Minify JS & CSS
11. Avoid redirects
12. Remove duplicate scripts
13. Configure Etags
14. Make AJAX cacheable

See the slides: https://stevesouders.com/docs/rich-web-experience-souders-theurer.ppt

Details on all the rules to follow

Rule 1: Make fewer HTTP requests

- Most browsers download only two resources at a time from a given hostname, as suggested in the HTTP/1.1 specification
  - However, some browsers open more than two connections per hostname
- To reduce the number of HTTP requests
  - Combine scripts
  - Combine style sheets
  - Combine images into an image map

```
<img usemap="map1" border=0 src="/images/imagemap.gif">
<map name="map1">
<area shape="rect" coords="0,0,31,31" href="home.html"
   title="Home">
. . .
</map>
```

  - Combine images using "**sprites**" (see next slide)
- Drawbacks
  - Images must be contiguous
  - Defining area coordinates is error prone

CSS Sprites

- An image sprite is a collection of images put into a single image
- Using images sprites reduces the number of server requests and saves bandwidth
- Consider img_navsprites.gif

which includes 3 separate images

The code

```
<!DOCTYPE html>
<html><head><style>
img.home  {  width:46px;  height:44px;
             background:url(img_navsprites.gif) 0 0;  }
img.next  { width:43px ;   height:44px;
             background:url(img_navsprites.gif) -91px 0;  }
</style></head>
<body>
<img class="home" src="img_trans.gif" width="1" height="1" />
<br><br>
<img class="next" src="img_trans.gif" width="1" height="1" />
</body></html>
```

- width:46px;height:44px; - Defines the portion of the image we want to use
- background:url(img_navsprites.gif) 0 0; - Defines the background image and its position (left 0px, top 0px)
- Check google.com with browser tools Network Net tab.

Rule 2: Use a CDN

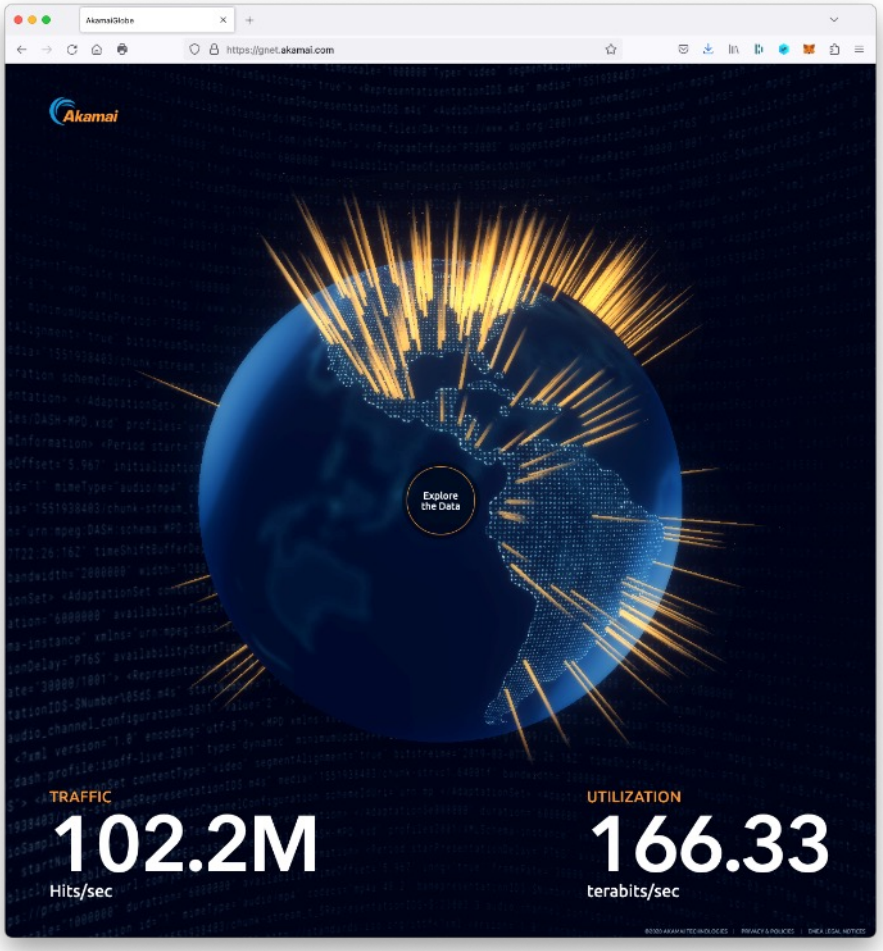
amazon.com	Akamai
aol.com	Akamai
cnn.com	cdn.turner.com
ebay.com	Akamai, Mirror Image
google.com	Google CDN
msn.com	SAVVIS
myspace.com	Akamai, Limelight
wikipedia.org	- not using CDN --
yahoo.com	Akamai
youtube.com	Google CDN, Akamai
apple.com	Akamai

- **Content Distribution Networks** have servers around the world
- They distribute your content so downloads can come from a **nearby** location
- Major CDN providers are
  - Akamai
    - Distributes MacOS and iOS updates
  - SAVVis
  - Limelight
  - OnApp
  - BityGravity
  - Amazon CloudFront
- Free CDNs [for individuals]
  - CloudFlare (+free Cert)
  - BootstrapCDN
  - Incapsula

More on Use a CDN: Akamai

Follow visualizing web application attacks

https://globe.akamai.com/



Rule 3: Add an Expires Header

- All caches use a set of rules to determine whether to deliver an object from the cache or request a new version
  - If the object's headers tell the cache not to keep the object, it won't
  - If the object is authenticated or secure, it won't be cached
  - A cached object is "fresh" (able to be sent to a client without checking with the server) if
    - It has an expiry time or other age-controlling directive that is set and still within the fresh period
    - If a browser cache has already seen the object and has been set to check once a session
    - If a proxy cache has seen the object recently, and it was modified long ago
  - If an object is "stale", the origin server will be asked to validate the object or tell the cache whether the copy that is has is still good
- As part of HTTP protocol there is a **Cache-Control** header
  - **Cache-Control** is an alternative to **Expires**
  - When **`cache-control: max-age`** is present, the response is stale if its current age is greater than the age value given (in seconds) at the time of a new request for the resource
  - The **`max-age`** directive on a response implies that the response is cacheable
- HTTP headers are sent by the server before the HTML, and only seen by the browser and any intermediate caches. Typical HTTP 1.1 response headers might look like this:

```
HTTP/1.1 200 OK
Date: Fri, 30 Oct 1998 13:19:41 GMT
Server: Apache/1.3.3 (Unix)
Cache-Control: max-age=3600, must-revalidate
Expires: Fri, 30 Oct 1998 14:19:41 GMT
Last-Modified: Mon, 29 Jun 1998 02:28:12 GMT
ETag: "3e86-410-3596fbbc"
Content-Length: 1040
Content-Type: text/html
```

More on Adding Expire Headers

- Here is a way to add an Expires header to files using the Apache httpd.conf file; add the lines:

```
<FilesMatch "\.(ico|pdf|flv|jpg|jpeg|png|gif|js|css|swf)$">
Header set Expires "Thu, 15 Apr 2020 20:00:00 GMT"
</FilesMatch>
```

- An Apache module enables / modifies expire headers:
- Apache Module mod_expire:
  - http://httpd.apache.org/docs/2.4/mod/mod_expires.html
  - This module controls the setting of the `Expires` HTTP header and the `max-age` directive of the Cache-Control HTTP header in server responses.
    `ExpiresDefault "access plus 1 month"`
  - `ExpiresByType image/gif "modification plus 5 hours 3 minutes"`
- For information on adding Expire headers and cache control to IIS 7, see:
  - http://www.iis.net/configreference/system.webserver/staticcontent/clientcache
- Nginx has an expires header module:
  - http://nginx.org/en/docs/http/ngx_http_headers_module.html

Rule 4: Gzip Components

	HTML	Scripts	Stylesheets
amazon.com	x		
aol.com	x	some	some
cnn.com			
ebay.com	x		
froogle.google.com	x	x	x
msn.com	x	deflate	deflate
myspace.com	x	x	x
wikipedia.org	x	x	x
yahoo.com	x	x	x
youtube.com	x	some	some

gzip scripts, stylesheets, XML, JSON (not images, PDF)

- Compression works when a web server like Apache is set up to "compress" resources and when a client's browser accepts such compressed resources.
- During the initial negotiation, if both browser and server support at least one "common" compression method (gzip, compress, etc) then the transfer is compressed.
- Client:

GET / HTTP/1.1

Accept-Encoding: gzip,deflate

- Server:

HTTP/1.1 200 OK

Vary: Accept-Encoding

Content-Encoding: gzip

- 99% of browsers support compression

More on Gzip'ing Components

- Apache Module mod_deflate (supports gzip and deflate):
  - http://httpd.apache.org/docs/2.0/mod/mod_deflate.html
  - Allows output from the server to be compressed before being sent to the client
  - `AddOutputFilterByType DEFLATE text/html text/plain text/xml`
- Apache Module mod_gzip:
  - http://sourceforge.net/projects/mod-gzip/
  - This is the older compression module for Apache
- See "Compressing Web Content with mod_gzip and mod_deflate":
  - http://www.linuxjournal.com/article/6802
- Microsoft "Using IIS Compression":
  - https://docs.microsoft.com/en-us/iis/extensions/iis-compression/using-iis-compression
- Compression with Nginx
  - To enable compression, use directive "gzip on"
  - By default, Nginx compresses only text/html
  - https://docs.nginx.com/nginx/admin-guide/web-server/compression/

Rule 5: Put Stylesheets at the top

- **Stylesheets block rendering in IE**; IE will examine all stylesheets before starting to render, so its best to
  - **Put stylesheets in the <head>**
- **Firefox** doesn't wait, it renders objects immediately and re-draws if it finds a stylesheet; that causes flashing during loading;
  - Put stylesheets in the <head>
- Use <link> (not @import)
  - There are two ways to load stylesheets
  - <link> includes the stylesheet in the web page

```
<link href="styles.css" type="text/css" />
```

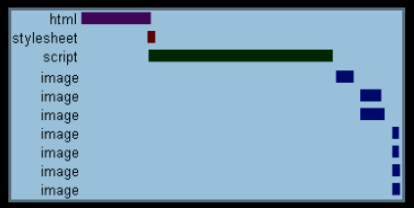
  - @import allows you to import one style sheet into another

```
<style type="text/css">@import url("styles.css");</style>
```

- General Rule for using @import
  - Link to a stylesheet for a specific page, but import a stylesheet that applies to all pages

Rule 6: Move Scripts to the Bottom

scripts block parallel downloads across all hostnames



scripts block rendering of everything below them in the page

`script defer` attribute is not a solution
- blocks rendering and downloads in FF
- slight blocking in IE

• As the **loading of JavaScript** can cause the browser to **stop rendering** the page until the JavaScript is fully loaded, one can avoid the delay by moving scripts to the bottom

• Example: move jQuery and Bootstrap libraries reference **right before </BODY>**

• A second option is the defer attribute

<script type="text/javascript" defer="defer"> *some script ....* </script>

•"defer" script attribute indicates that the script is not going to generate any document content. The browser can continue parsing and drawing the page

https://developer.mozilla.org/en-US/docs/Web/HTML/Element/script
And scroll down to "defer"

Rule 7: Avoid CSS Expressions (obsolete)

- Used to set CSS properties dynamically in IE

```
Width: expression(document.body.clientWidth < 600 ? “600px” : “auto”);
```

- Problem: expression may execute many times
  - Mouse moves, key press, resize, scroll, etc
- Alternatives
  - One-time expressions
  - Event handlers
- Expression overwrites itself
- **Note: no longer relevant, as fixed in later versions of IE**

```
<style>
P { background-color: expression(altBgcolor(this)); }
</style>
<script>
function altBgcolor(elem) {
   elem.style.backgroundColor =
(new Date()).getHours()%2 ? “F08A00” : “#B8D4FF”; }
</script>
```

Rule 8: Make JS and CSS External

- JavaScript can be placed inline, e.g

```
<script type="text/javascript>var foo = "bar"; </script>
```

- Or as an external script, e.g.

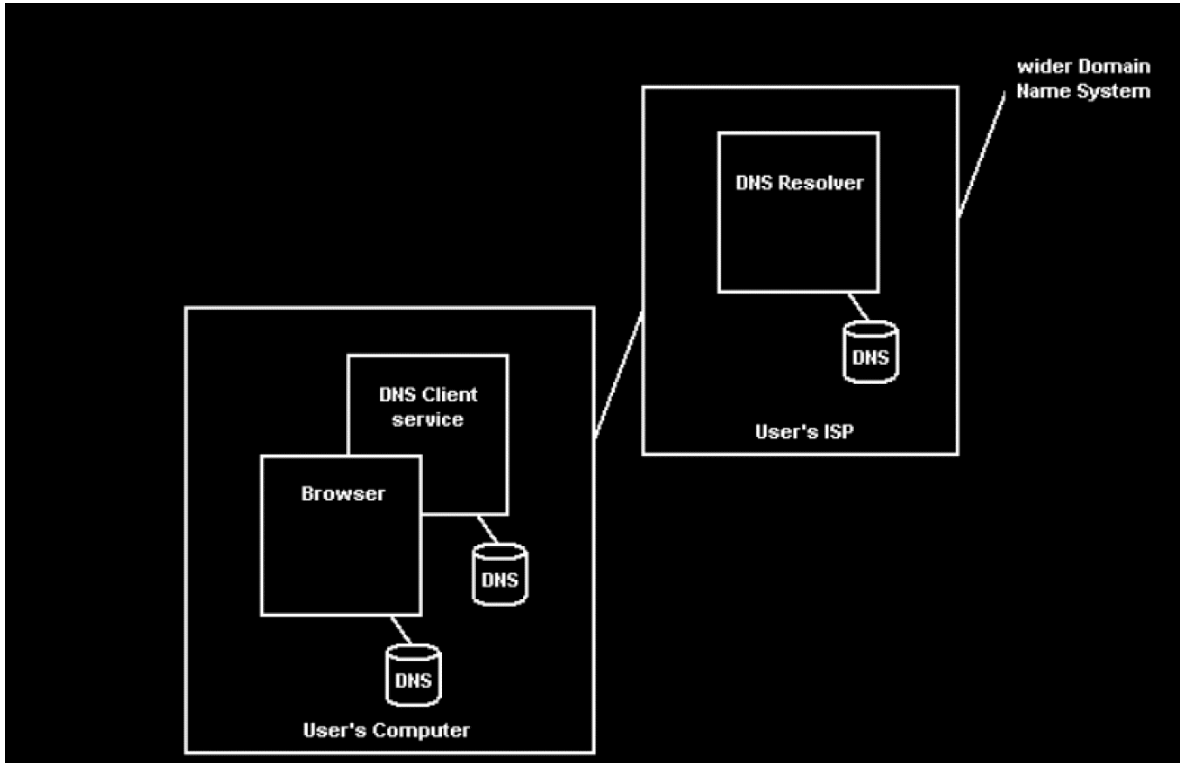
```
<script src="foo.js" type="text/javascript"></script>
```

- Placing JavaScript and CSS inline makes the document bigger
- Making JavaScript and CSS external implies more HTTP requests, but
  - As HTML documents are not typically cached, so inlining JavaScript code will cause the same bytes to be downloaded on every page view
  - **External JS and CSS can be cached**
- Defining JavaScript externally is typically better
- Download external files after onload

```
window.onload = downloadComponents;
function downloadComponents ( ) {
   var elem = document.createElement("script");
   elem.src = http://.../file1.js;
   document.body.appendChild(elem);
   . . . . .
}
```

Rule 9: Reduce DNS lookups

- Typically, each look up takes 20 – 120 milliseconds
- DNS lookups will block parallel downloads
- The operating system and the browser both have DNS caches
- As a general rule it is best to reduce the number of unique hostnames used in a web page



Rule 10: Minify JavaScript

	Minify External?	Minify Inline?
www.amazon.com	no	no
www.aol.com	no	no
www.cnn.com	no	no
www.ebay.com	yes	no
froogle.google.com	yes	yes
www.msn.com	yes	yes
www.myspace.com	no	no
www.wikipedia.org	no	no
www.yahoo.com	yes	yes
www.youtube.com	no	no

minify inline scripts, too

- Minification, or minify, is the process of removing all unnecessary characters from source code, without changing its functionality
  - **Unnecessary characters** usually include white space characters, new line characters, comments , block delimiters used to add readability to code, but are not required for execution
- There are many programs that minify JavaScript code, see

  http://en.wikipedia.org/wiki/Minify

  (JSmin, packer, Google Closure Compiler)
- See this website for example of minification (Google analytics.js): purecss.io

Minified JavaScript

- Example of "minified code" is the Google Maps engine at:

  http://maps.google.com/
- An example of "minified library" is Bootstrap at:
  https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/js/bootstrap.**min.js**
- You can use PHP code from Google (HTTP server for minification) to do the job of minifying CSS & JavaScript :

  http://code.google.com/p/minify/
- The original JSMin minifier from Crockford:

  http://www.crockford.com/javascript/jsmin.html
- An on-the-fly minifier of JavaScript/CSS for IIS can be found here:

  http://highoncoding.com/Articles/777_Minify_CSS_and_JavaScript_in_ASP_NET.aspx

Minify vs. Obfuscate

An obfuscator minifies but also makes modifications to the program, changing variable names, function names, etc. making it much harder to understand; JSMin and Dojo are two minifiers

	Original	JSMin Savings	Dojo Savings
www.amazon.com	204K	31K (15%)	48K (24%)
www.aol.com	44K	4K (10%)	4K (10%)
www.cnn.com	98K	19K (20%)	24K (25%)
www.myspace.com	88K	23K (27%)	24K (28%)
www.wikipedia.org	42K	14K (34%)	16K (38%)
www.youtube.com	34K	8K (22%)	10K (29%)
Average	**85K**	**17K (21%)**	**21K (25%)**

minify – it's safer

Rule 11: Avoid Redirects

- Redirects are used to map users from one URL to another
  - However, redirects insert an extra HTTP round-trip between user and origin server
- **PHP Redirect**

```
<?
Header( "HTTP/1.1 301 Moved Permanently" );
Header( "Location: http://www.new-url.com" );
?>
```

- **JSP (Java) Redirect**

```
<%
response.setStatus(301);
response.setHeader( "Location", "http://www.new-url.com/" );
response.setHeader( "Connection", "close" );
%>
```

- **CGI PERL Redirect**

```
$q = new CGI;
print $q->redirect("http://www.new-url.com/");
```

- **Redirect Old domain to New domain (htaccess redirect)**
  - Create a .htaccess file with the below code, it will ensure that all your directories and pages of your old domain will get correctly redirected to your new domain.
    The .htaccess file needs to be placed in the root directory of your old website (i.e the same directory where your index file is placed)

```
Options +FollowSymLinks
    RewriteEngine on
    RewriteRule (.*) http://www.newdomain.com/$1 [R=301,L]
```

  - REPLACE www.newdomain.com in the above code with your actual domain name.
  - contact every backlinking site to modify their backlink to point to your new website.
  - **Note*** This .htaccess method of redirection works ONLY on Linux servers having the Apache Mod-Rewrite module enabled.

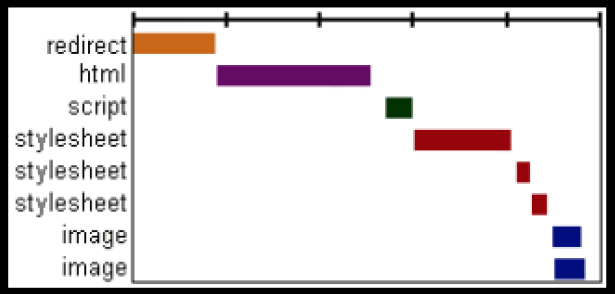
Rule 11: Avoid Redirects

3xx status codes - mostly 301 and 302

```
HTTP/1.1 301 Moved Permanently
Location: http://stevesouders.com/newuri
```

add Expires headers to cache redirects

worst form of blocking



Rule 12: Remove Duplicate Scripts

- Hurts performance
  - Extra HTTP requests
  - Extra executions
- Is it atypical?
  - 2 of the top 10 websites contain duplicate scripts (JavaScript and CSS)

Rule 13: Configure ETags

- Etags are used by clients and servers to verify that a cached resource is valid
  - To check that the resource (image, script, stylesheet, etc.) in the browser's cache matches the one on the server
  - If there is a match, the server returns a 304
- Unique identifier returned in response

```
Etag: "c8897e-aee-4165acf0"
Last-Modified: Thu, 07 Oct 2008 20:54:08 GMT
```

- Used in conditional GET requests

```
If-None-Match: "c8897e-aee-4165acf0"
If-Modified-Since: Thu, 07 Oct 2008 20:54:08 GMT
```

- If Etag doesn't match, can't send 304
- Etag format varies across web servers
  - `Apache: inode-size-timestamp`
  - `IIS: FileTimestamp:ChangeNumber`
- Use 'em or lose 'em
  - `Apache: FileETag none`
  - `IIS: http://support.microsoft.com/kb/922703/`
  - `http://stackoverflow.com/questions/477913/how-do-i-remove-etag-headers-from-iis7`

Rule 14: Make AJAX Cacheable and small

- The URL of an AJAX request is included inside the HTML; as it is not bookmarked or linked to, the requesting page can be cached by the browser
- The AJAX request URL should include a dynamic variable, e.g., dateAndtime, so if the page is changed at the server, the new page will be downloaded
- As long as the AJAX request page has not changed, e.g., an address book would change infrequently, it is best to have the browser cache it
- See http://blog.httpwatch.com/2009/08/07/ajax-caching-two-important-facts/ for more details. In general, use of these response headers that make AJAX response cacheable:
  - Expires, Last-Modified, Cache-Control
  - Example: stock quote that updates every 10 seconds

```
GET /yab/[…]&r=0.5289571 HTTP/1.1
Host: us.xxx.mail.yahoo.com

HTTP/1.1 200 OK
Date: thu, 12 Apr 2007 19:39:09 GMT
Cache-Control: private, max-age=0
Last-Modified: Sat, 31 Mar 2007 01:17:17 GMT
Content-type: text/xml; charset=utf-8
Content-Encoding: gzip
```

Updated Yahoo Best Practices (2011)

Yahoo updated their list of best practices in 2011. See:

http://developer.yahoo.com/performance/rules.html

1.Flush the buffer early
2.Use GET for AJAX requests
3.Post-load components
4.Preload Components
5.Reduce the number of DOM Elements
6.Split Components Across Domains
7.Minimize the number of iframes
8.No 404s
9.Reduce cookie size
10.Use cookie-free domains for components
11.Minimize DOM access
12.Develop smart event handlers
13.Avoid filters
14.Optimize images
15.Optimize CSS sprites
16.Don't scale images in html
17.Make favicon.ico small and cacheable
18.Keep components under 25K
19.Pack components into a multipart document
20.Avoid Empty image src

Some New Rules (2011)

- **Avoid empty src or href**

You may expect a browser to do nothing when it encounters an empty image src. However, it is not the case in most browsers. IE makes a request to the directory in which the page is located; Safari, Chrome, Firefox make a request to the actual page itself. This behavior could possibly corrupt user data, waste server computing cycles generating a page that will never be viewed, and in the worst case, cripple your servers by sending a large amount of unexpected traffic.

- **Use GET for AJAX requests**

When using the XMLHttpRequest object, the browser implements POST in two steps: (1) send the headers, and (2) send the data. It is better to use GET instead of POST since GET sends the headers and the data together (unless there are many cookies). IE's maximum URL length is 2 KB, so if you are sending more than this amount of data you may not be able to use GET.

- **Reduce the number of DOM elements**

A complex page means more bytes to download, and it also means slower DOM access in JavaScript. Reduce the number of DOM elements on the page to improve performance.

Some New Rules (cont'd)

- **Avoid HTTP 404 (Not Found) error**

Making an HTTP request and receiving a 404 (Not Found) error is expensive and degrades the user experience. Some sites have helpful 404 messages (for example, "Did you mean ...?"), which may assist the user, but server resources are still wasted.

- **Reduce cookie size**

HTTP cookies are used for authentication, personalization, and other purposes. Cookie information is exchanged in the HTTP headers between web servers and the browser, so keeping the cookie size small minimizes the impact on response time.

- **Use cookie-free domains**

When the browser requests a static image and sends cookies with the request, the server ignores the cookies. These cookies are unnecessary network traffic. Make sure that static component requests are cookie-free (i.e., use static.mydomain.com to serve static content).

- **Do not scale images in HTML**

Web page designers sometimes set image dimensions by using the width and height attributes of the HTML image element. Avoid doing this since it can result in images being larger than needed. For example, if your page requires image myimg.jpg which has dimensions 240x720 but displays it with dimensions 120x360 using the width and height attributes, then the browser will download an image that is larger than necessary. (This rule conflicts with Responsive Web Design patterns)

Some New Rules (cont'd)

- **Make favicon small and cacheable**

A favicon is an icon associated with a web page; this icon resides in the favicon.ico file in the server's root. Since the browser requests this file, it needs to be present; if it is missing, the browser returns a 404 error (see "Avoid HTTP 404 (Not Found) error" above). Since favicon.ico resides in the server's root, each time the browser requests this file, the cookies for the server's root are sent. Making the favicon small and reducing the cookie size for the server's root cookies improves performance for retrieving the favicon. Making favicon.ico cacheable avoids frequent requests for it. Set expires header to a few months into the future, and limit size to 1K.

Some New Rules (cont'd)

- Load only glyphs that are used when loading a font

- The majority of fonts support different languages – some may even have 20,000+ glyphs (with a size of 7-8MB!).
- If you need to support only a subset – say, English alphabet + characters you can save 7 MB by requesting only a specific subset:
  - Google Fonts:
    - &subset=latin
    - &text=onlyglyphsyouneed
  - Run specialized software to prune unneeded characters
  - CSS Unicode-range:

```
unicode-range: U+0025-00FF, U+4??; /* multiple values */
```

https://developer.mozilla.org/en-US/docs/Web/CSS/@font-face/unicode-range

Updated Yahoo Best Practices (2012)

1. **Minimize HTTP Requests**
2. **Use a Content Delivery Network**
3. **Add an Expires or a Cache-Control Header**
4. **Gzip Components**
5. **Put Stylesheets at the Top**
6. **Put Scripts at the Bottom**
7. **Avoid CSS Expressions**
8. **Make JavaScript and CSS External**
9. **Reduce DNS Lookups**
10. **Minify JavaScript and CSS**
11. **Avoid Redirects**
12. **Remove Duplicate Scripts**
13. **Configure Etags**
14. **Make Ajax Cacheable**
15. **Flush the Buffer Early**
16. **Use GET for AJAX Requests**
17. **Post-load Components**
18. **Preload Components**
19. **Reduce the Number of DOM Elements**
20. **Split Components Across Domains**
21. **Minimize the Number of iframes**
22. **No 404s**
23. **Reduce Cookie Size**
24. **Use Cookie-free Domains for Components**
25. **Minimize DOM Access**
26. **Develop Smart Event Handlers**
27. **Choose <link> over @import**
28. **Avoid Filters**
29. **Optimize Images**
30. **Optimize CSS Sprites**
31. **Don't Scale Images in HTML**
32. **Make favicon.ico Small and Cacheable**
33. **Keep Components under 25K**
34. **Pack Components into a Multipart Document**
35. **Avoid Empty Image src**

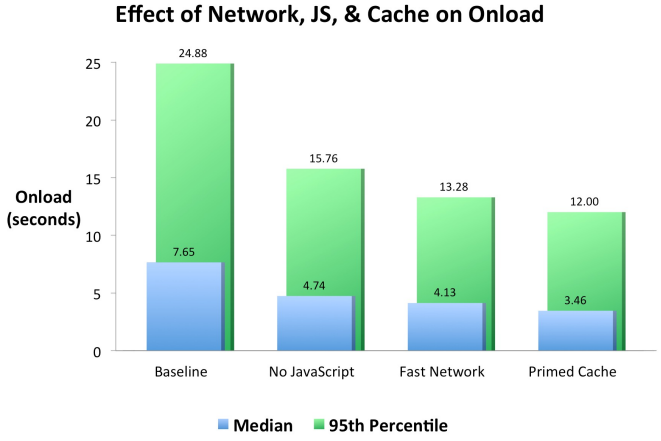
Only 23 rules could be tested with YSlow (no longer available)

New Rule: Cache is King

- In a blog in 2012, Steve Souders commented that the keys to fast Web 2.0 apps are the following:
  - **Using Ajax** reduces network traffic to just a few JSON requests.
  - **Optimizing JavaScript** (downloading scripts asynchronously, grouping DOM modifications, yielding to the UI thread, etc.) allows requests to happen in parallel and makes rendering happen sooner.
  - **Better caching** means many of the web app's resources are stored locally and don't require any HTTP requests.
- In particular:
  - **Using Ajax**, for example, doesn't make the initial page load time much faster (and often makes it slower if you're not careful), but subsequent "pages" (user actions) are snappier.
  - **Optimizing JavaScript,** on the other hand, makes both the first page view and subsequent pages faster.
  - **Better caching** sits in the middle: The very first visit to a site isn't faster, but subsequent page views are faster. Also, even after closing their browser the user gets a faster initial page when she returns to your site – so the performance benefit transcends browser sessions.
- See: https://www.stevesouders.com/blog/page/6/

New Rule: Cache is King (cont'd)

- Which has the biggest impact? Steve Souders run tests with these scenarios:
  - **Baseline** – Run the Alexa Top 1000 through WebPagetest using a simulated DSL connection speed (1.5 Mbps down, 384 Kbps up, 50ms RTT). Each URL is loaded three times and the median (based on page load time) is the final result. We only look at the "first view" (empty cache) page load.
  - **Fast Network** – Same as Baseline except use a simulated FIOS connection: 20 Mbps down, 5 Mbps up.
  - **No JavaScript** – Same as Baseline except use the new "noscript" option to the RESTful API. This is the same as choosing the browser option to disable JavaScript.
  - **Primed Cache** – Same as Baseline except only look at "repeat view". This test looks at the best-case scenario for the benefits of caching given the caching headers in place today. Since not everything is cacheable, some network traffic still occurs.



- Result: Primed cache is fastest, Fast Network is second, no JavaScript is third.

New Rule: Cache is King (cont’d)

- What is the cache size in the top browsers?
  - **Google Chrome**: Chrome typically allocates a portion of your computer's disk space for its cache. The specific size can vary based on your browsing habits and the available disk space on your device. By default, Chrome doesn't have a fixed cache size limit and may dynamically adjust it as needed.
  - **Mozilla Firefox**: Firefox also dynamically manages its cache size based on available disk space and your browsing activities. However, you can manually adjust the cache size in Firefox settings. To do this, type "about:config" in the address bar, accept the risk, and search for "browser.cache.disk.capacity". You can then set the value (in bytes) to your preferred size.
  - **Safari**: Safari, being the default browser for macOS and iOS devices, manages its cache similarly. The cache size can vary, but Safari typically optimizes it based on available disk space and browsing patterns. Users generally don't have direct control over the cache size in Safari settings.

Performance checklist

1. Reduce number of HTTP requests
   combine resources, use sprites, etc.

   Use bundler plugins

2. Minify & compress resources
   (css, js, svg, etc)

   Use bundlers (vite, webpack)

3. Use a CDN and HTTP2/3

4. Utilize cache
   etag, expire headers

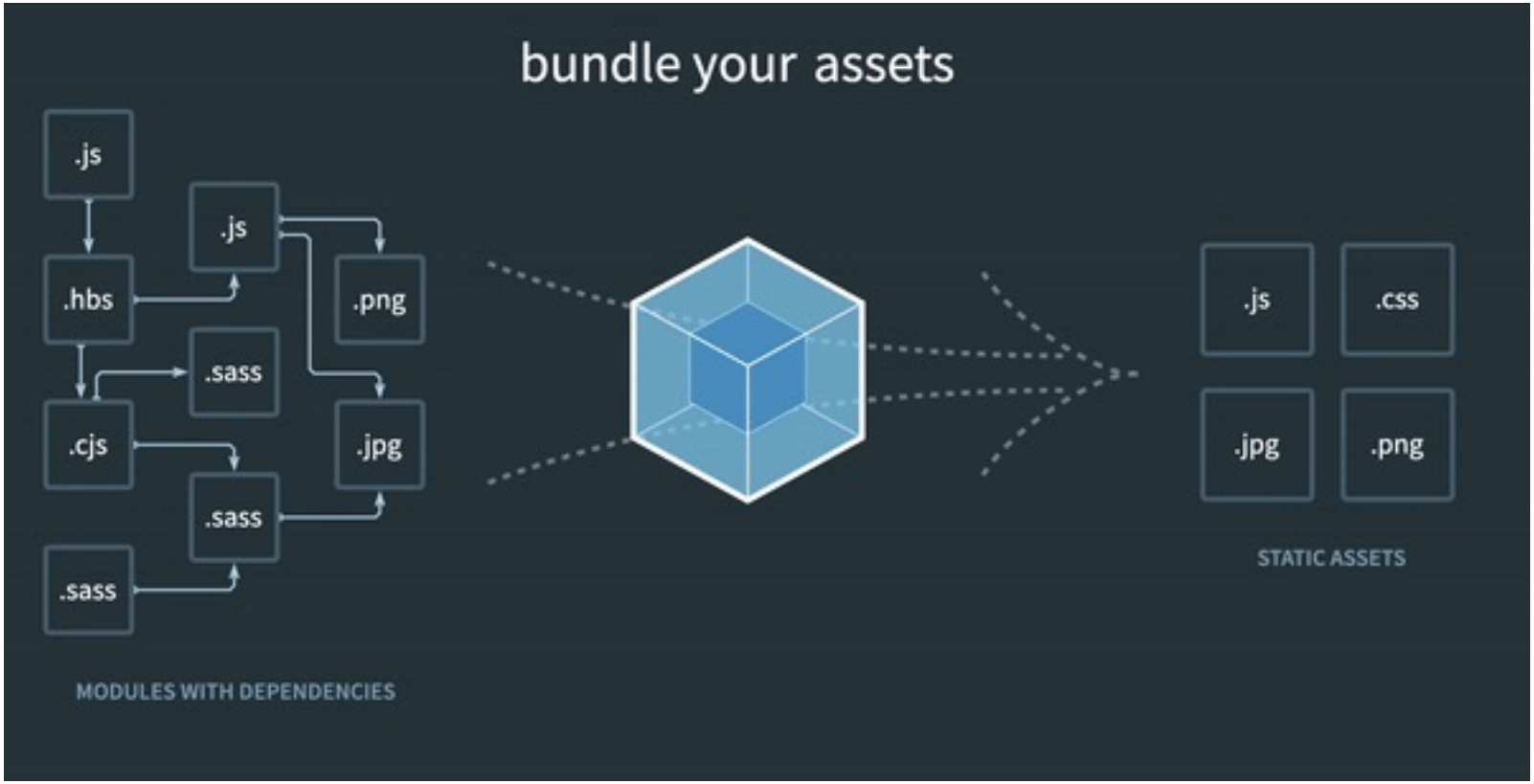
   Configure your server

5. Avoid redirects

6. Defer loading or lazy-load less important resources
   use <script async/defer> or dynamic: await import()

7. Limit characters of the font you are importing

Bundlers

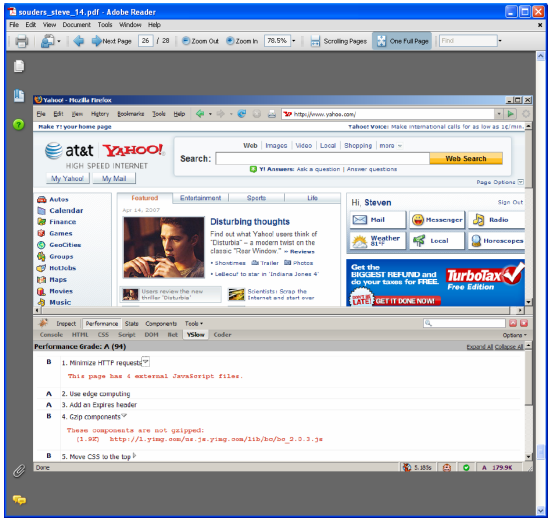


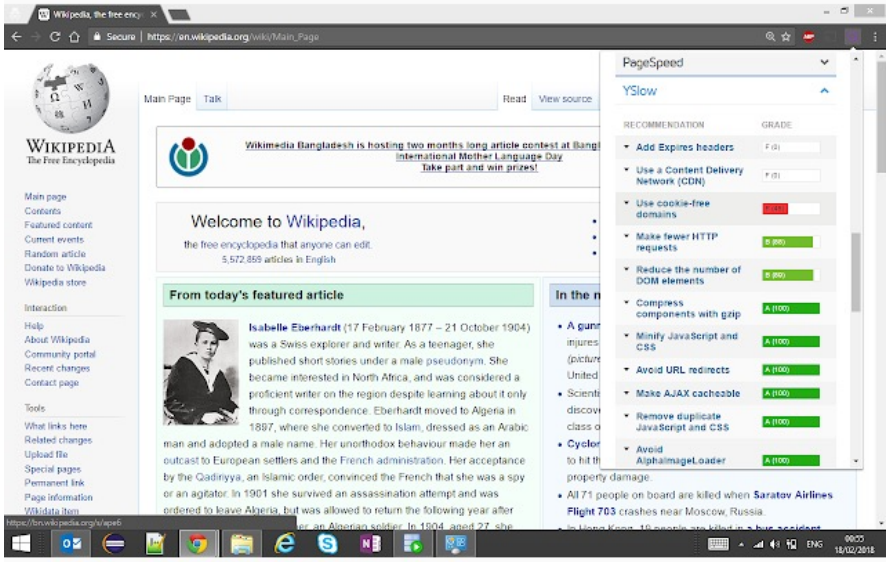
Bundlers - Vite

A complete modern frontend tool (https://vite.dev/ , free, open source)

- Can automatically build the dependency graph and chunk necessary files
  - Example: 100 js files -> 1 entrypoint + 3 chunks; 100 css files -> 1 css chunk
- Transforms JS – UMD/ESM
- Supports web-workers, web-assembly, jsx, json imports, CSP
- Works with JS/TS and Frameworks:
  - Vue (Native), React, Angular, Svelte, etc…
- Support plugins:
  - minification, image compression/embedding, etc…
- Supports live updates (HMR)
- Has a built-in server proxy
- Based on esbuild, significantly faster than webpack
- And many more

Yslow (obsolete)





- http://developer.yahoo.com/yslow (retired)
- Grades web pages for each rule described earlier
- Firefox add-on
- Minimize HTTP requests
- Add an expires header
- Gzip components
- Tests 23 rules (Yslow V2) or original 14 rules (Classic V1)
- Original version stopped working with Firefox 46.

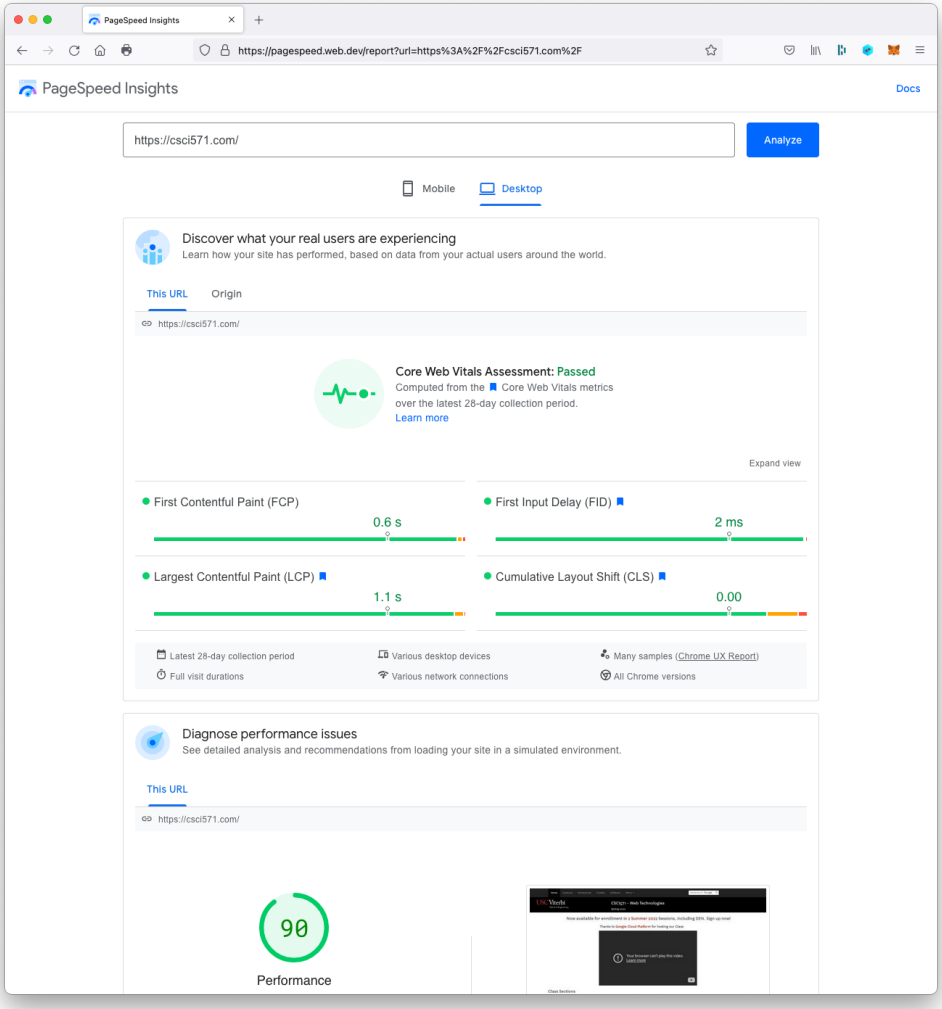
Google's PageSpeed Insights (PSI)

- Google offers a similar service to Yahoo's YSlow called PageSpeed Insights. See:

  http://developers.google.com/speed
- Enter the URL to test at:

  https://pagespeed.web.dev/
- Page Speed is also available in Google Analytics
- PageSpeed can integrate with **Apache** and **Nginx** web servers to automatically optimize your site

  https://developers.google.com/speed/pagespeed/module



PageSpeed Insights Basics

- PageSpeed Insights (PSI) reports on the user experience of a page on both mobile and desktop devices and provides suggestions on how that page may be improved.
- PSI provides both lab and field data about a page.
- **Lab data** is performance data collected within a controlled environment with predefined device and network settings.
  - Strengths
    - Helpful for debugging performance issues
    - End-to-end and deep visibility into the UX
    - Reproducible testing and debugging environment
  - Limitations
    - Might not capture real-world bottlenecks
    - Cannot correlate against real-world page KPIs
- **Field data** is performance data collected from real page loads your users are experiencing in the wild.
  - Strengths
    - Captures true real-world user experience
    - Enables correlation to business key performance indicators
  - Limitations
    - Restricted set of metrics
    - Limited debugging capabilities

PageSpeed Insights Basics (cont' d)

- PSI reports real users':
  - **First Contentful Paint (FCP),** an important, user-centric metric for measuring perceived load speed. FCP measures the time from when the user first navigates to the page to when any part of the page's content is rendered on the screen.
  - **Interaction to Next Paint (INP)**, a stable Core Web Vital metric that assesses page responsiveness, observing the latency of all click, tap, and keyboard interactions with a page throughout its lifespan.
  - **Largest Contentful Paint (LCP)**, a stable Core Web Vital metric for measuring perceived load speed. It marks the point in the page load timeline when the page's main content has likely loaded.
  - **Cumulative Layout Shift (CLS)**, a user-centric metric for measuring visual stability because it helps quantify how often users experience unexpected layout shifts. Unexpected layout shifts can disrupt the user experience.
  - **Time to First Byte (TTFB)**, a foundational metric for measuring connection setup time and web server responsiveness in both the lab and the field. It measures the time between the request for a resource and when the first byte of a response begins to arrive.
- PSI classifies the quality of user experiences into three buckets: **Good**, **Needs Improvement**, or **Poor**.

	Good	Needs Improvement	Poor
FCP	[0, 1800ms]	(1800ms, 3000ms]	over 3000ms
LCP	[0, 2500ms]	(2500ms, 4000ms]	over 4000ms
CLS	[0, 0.1]	(0.1, 0.25]	over 0.25
INP	[0, 200ms]	(200ms, 500ms]	over 500ms
TTFB (experimental)	[0, 800ms]	(800ms, 1800ms]	over 1800ms

PageSpeed Insights Basics (cont' d)

PSI diagnostics reports recommendations on all of the following:

- Minimize main-thread work
- Largest Contentful Paint element
- Reduce JavaScript execution time
- Reduce the impact of third-party code
- Eliminate render-blocking resources
- Minify CSS
- Reduce unused CSS
- Avoid an excessive DOM size
- Defer offscreen images
- Reduce unused JavaScript
- Preload Largest Contentful Paint image
- Image elements do not have explicit width and height
- JavaScript execution time
- Properly size images
- Minify JavaScript
- Efficiently encode images
- Preconnect to required origins
- Avoid multiple page redirects
- Preload key requests
- Use video formats for animated content
- Serve images in next-gen formats
- Enable text compression
- Serve static assets with an efficient cache policy
- Ensure text remains visible during webfont load
- Properly size images
- Avoid serving legacy JavaScript to modern browsers
- Avoid long main-thread tasks
- Avoid large layout shifts
- Initial server response time
- Avoids enormous network payloads
- Avoid chaining critical requests
- Minimize third-party usage
- Remove duplicate modules in JavaScript bundles
- User Timing marks and measures
- Lazy load third-party resources with facades
- Largest Contentful Paint image was not lazily loaded
- Avoids document.write()
- Avoid non-composited animations

http archive – State of the Web

- This report captures a long view of the web, including the adoption of techniques for efficient network utilization and usage of web standards like HTTPS.

  https://httparchive.org/reports/state-of-the-web